Lecture 7 recap

In Lecture 7:

- Revised power series, including Taylor series representations of functions.
- Saw an example of how to solve an ODE using a power series.
- Revised the ratio test to determine the radius of convergence of a series.

Lecture 8

In Lecture 8, we will:

1. Continue investigating convergence of power series
2. Learn the power series solution methodology
3. Solve the Airy differential equation

Exercise 2.6: What does a given power series converge to?

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$c_n = 1/n!$

$c_{n+1} = 1/(n+1)!$

Use the ratio test to calculate the radius of convergence

$$\lim_{n\to\infty} \left| \frac{n!\, t}{(n+1)!} \right|$$

$$\lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} t \right| < 1$$

missing t added

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$$\lim_{n\to\infty} \left| \frac{n!\,t}{(n+1)!} \right| = \lim_{n\to\infty} \left| \frac{\cancel{n!}\,t}{n+1} \right|$$

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Therefore, the series converges for all t (infinite radius of convergence).

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$m = n - 1$

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Also, note that

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Thus, we have the IVP

$f_0(t_1) = -1$

$$P'(t) - P(t) = 0 \quad \text{with} \quad P(0) = 1$$

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From Topic 1, we know the IVP has a unique solution, which is $P(t) = \mathrm{e}^t$.

Thus, we have the limit of the series.

Exercise 2.7: Taylor series don't always converge to the original function

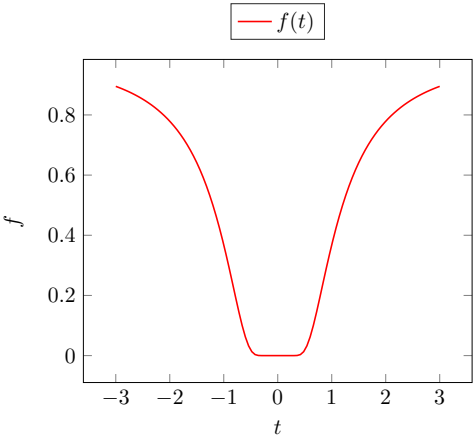
Consider the function $f(t) = \begin{cases} e^{-\frac{1}{t^2}} & t \neq 0 \\ 0 & t = 0 \end{cases}$

$f(t) = e^{-u} \qquad u = t^{-2}$

$df/dt = df/du \; du/dt$

$= -e^{-u} \times -2t^{-3}$

$= 2t^{-3} e^{-1/t^2}$



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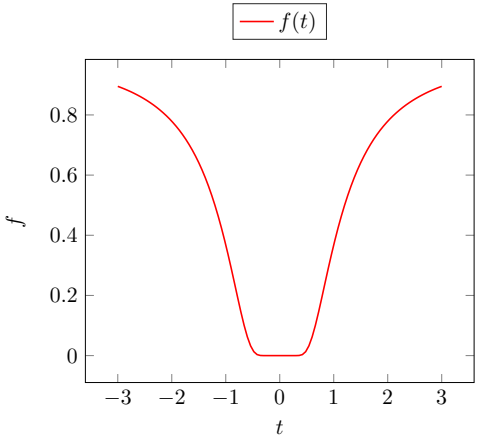

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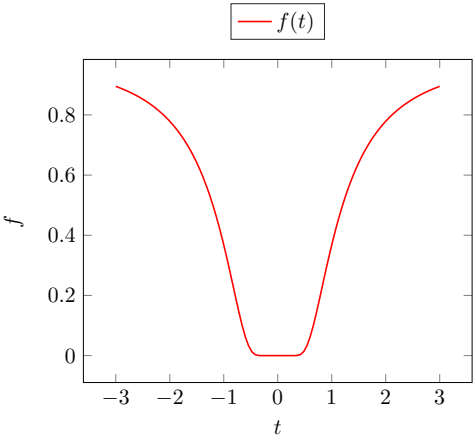

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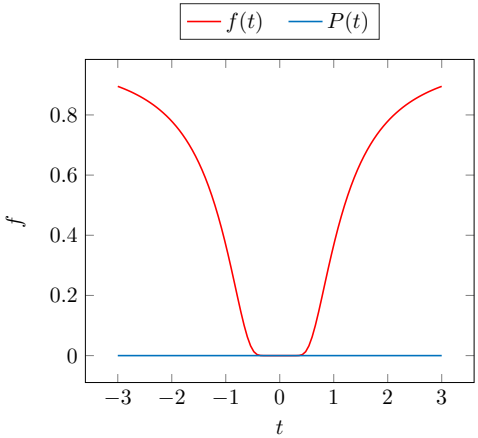
- $f(t)$ is continuous and infinitely differentiable for all t
- $f^{(n)}(0) = 0$ for all n
- Thus, the Taylor series of $f(t)$ is

$$P(t) = \sum_{n=0}^{\infty} \frac{0}{n!} t^n = 0 \quad \text{for all } t$$



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4. Solve for the coefficients using recurrence relations.
5. Check if you have the right number of linearly independent solutions.

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$$\sum_{n=0}^{\infty} c_n n(n-1)t^{n-2} + \sum_{n=0}^{\infty} c_n t^n = 0$$

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$$\underbrace{\sum_{m=-2}^{\infty} c_{m+2}(m+2)(m+1)t^m}_{= 0 \text{ for } m = -2, -1} + \sum_{m=0}^{\infty} c_m t^m = 0$$

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We require

$$c_{m+2}(m+2)(m+1) + c_m = 0 \quad \Rightarrow \quad c_{m+2} = \frac{-c_m}{(m+2)(m+1)} \quad \text{for} \quad m = 0, 1, 2, \ldots$$

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We have found two independent series solutions

$$c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} t^{2n} = c_0 \left(1 - \frac{t^2}{2!} + \dots\right)$$

$$\text{and} \quad c_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} t^{2n+1} = c_1 \left(t - \frac{t^3}{3!} + \dots\right)$$

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Replace c_0 and c_1 with C_1 and C_2, then the general solution is

$$\begin{aligned} y(t) &= C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} t^{2n} + C_2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} t^{2n+1} \\ &= C_1 \cos(t) + C_2 \sin(t) \end{aligned}$$

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Airy differential equation

- Is the 2nd-order, homogeneous linear ODE

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$$\sum_{n=0}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Use $n - 2 \to m + 1$ in first sum

$$\Rightarrow \quad \sum_{m=-3}^{\infty} c_{m+3}(m+3)(m+2)x^{m+1} - \sum_{m=0}^{\infty} c_m x^{m+1} = 0$$

$n-2 = m+1$

if $n=0$ then $m=-3$

$$\Rightarrow \quad \sum_{m=-3}^{\infty} c_{m+3}(m+3)(m+2)x^{m+1} - \sum_{m=0}^{\infty} c_m x^{m+1} = 0$$

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$$c_3 = \frac{1}{2 \cdot 3} c_0$$

$$c_4 = \frac{1}{3 \cdot 4} c_1$$

$$\Rightarrow \quad \sum_{m=-3}^{\infty} c_{m+3}(m+3)(m+2)x^{m+1} - \sum_{m=0}^{\infty} c_m x^{m+1} = 0$$

$$\Rightarrow 2c_2 + \sum_{m=0}^{\infty} \left\{ c_{m+3}(m+3)(m+2) - c_m \right\} x^{m+1} = 0$$

$$\Rightarrow \quad c_2 = 0 \qquad \text{and} \qquad c_{m+3} = \frac{c_m}{(m+3)(m+2)}$$

$$c_3 = \frac{1}{2 \cdot 3} c_0, \quad \ldots, \quad c_{3k} = \frac{1}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3k-1) \cdot 3k} c_0$$

$$c_4 = \frac{1}{3 \cdot 4} c_1, \quad \ldots, \quad c_{3k+1} = \frac{1}{3 \cdot 4 \cdot 6 \cdot 7 \cdots 3k \cdot (3k+1)} c_1$$

Two independent solutions:

$$y_1(x) = 1 + \sum_{k=1}^{\infty} \frac{x^{3k}}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3k-1) \cdot 3k}$$

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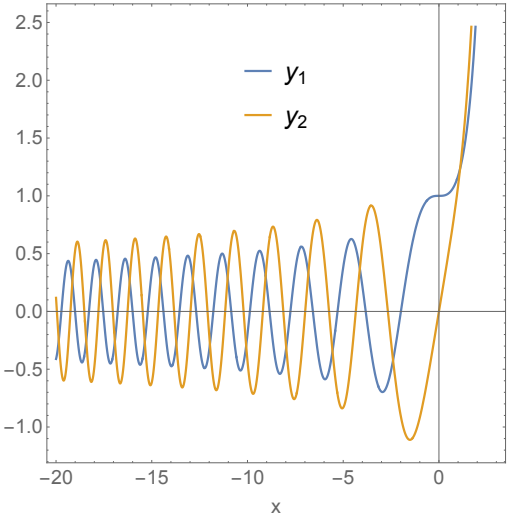

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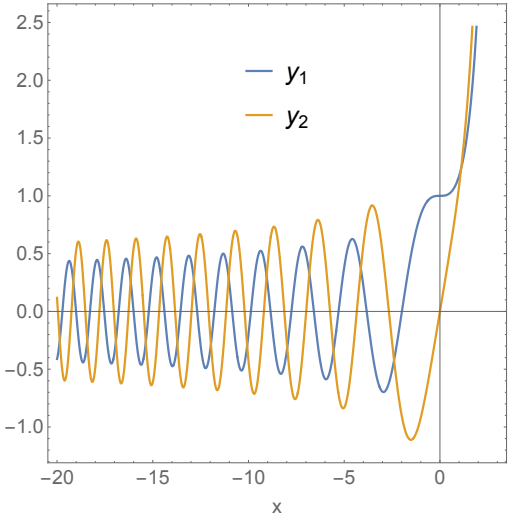

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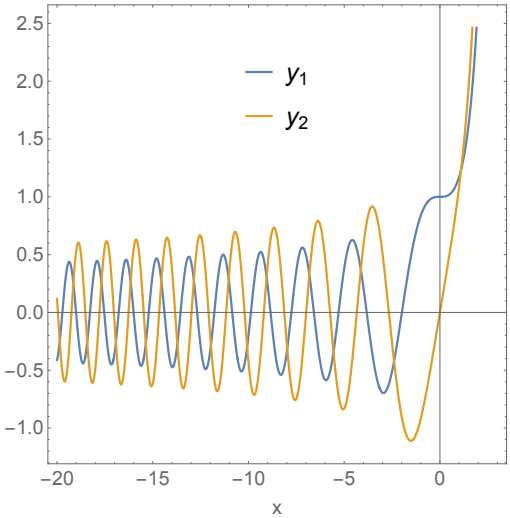
The general solution of Airy's eqn.

$$y(x) = C_1\, y_1(x) + C_2\, y_2(x)$$



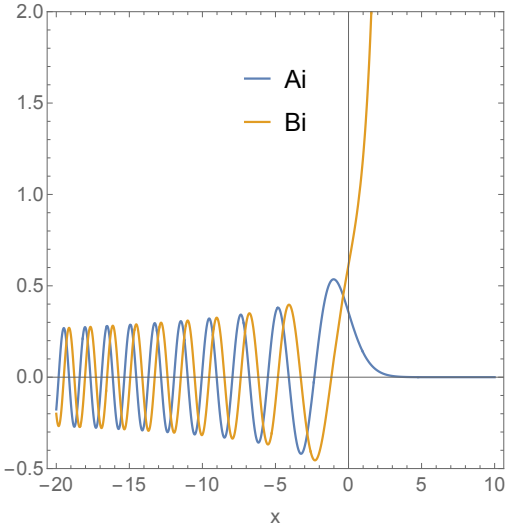
Airy functions

The *Airy functions* are

$$\mathrm{Ai}(x) = \alpha\big[y_1(x) - \beta y_2(x)\big]$$

$$\mathrm{Bi}(x) = \sqrt{3}\alpha\big[y_1(x) + \beta y_2(x)\big]$$

where $\alpha \approx 0.355$ and $\beta \approx 0.729$.



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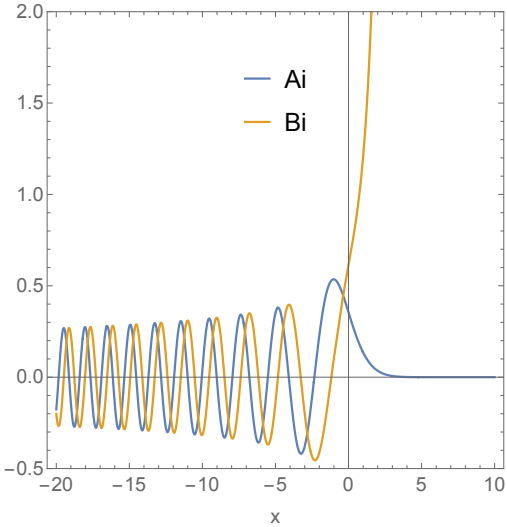

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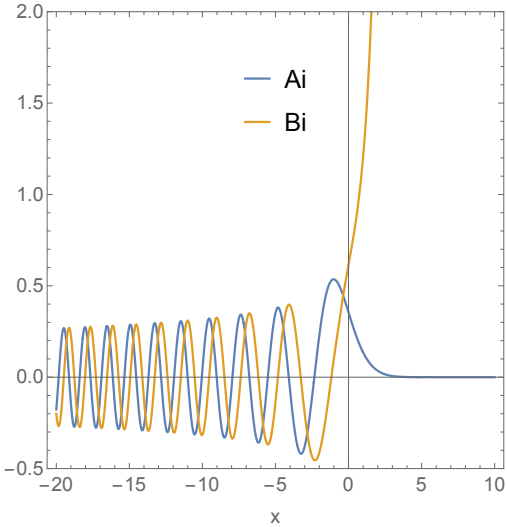

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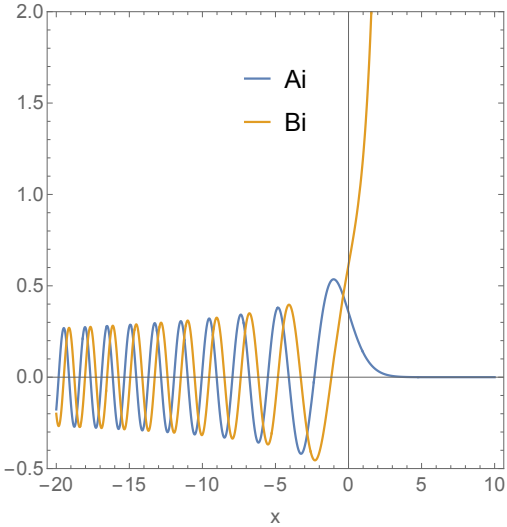
We'll see $\mathrm{Ai}(x)$ again in Topic 6.